

A Brief History of Illumination

From the sun to color changing LED strips

The Sun

Free heat and light, but only available for about half the day. Doesn't work well when it's cloudy. Really doesn't work well during winter, particularly in extreme northern and southern latitudes.

Available since the dawn of the solar system.

Fire

Good for cooking and lighting. Can be used at night. Requires a fuel source and knowledge of how to start it ablaze. Candles made fire more portable.

Available since early humans got the knack for it.

Incandescent Bulbs

Convenient source of light, easily switched on and off. Heat is a byproduct, but usually considered not a very useful one. Requires a power source and wiring.

Available commercially since the 1880s. Grew to be the standard light source in most homes, cars, and portable lighting (like flashlights.)

Red and Green LEDs

Rose to prominence in the late 1970s. Though only red, green, and yellow (created by mixing red and green) were available.

LEDs represented a more efficient source of light with less waste heat produced. Not a substitute for incandescent bulbs, however, because of limited colors available and lower overall brightness.

Commercially popular for visual feedback, like power and status indicators. Also, LEDs arranged in figure eight patterns formed the displays of calculators, digital clocks and early digital watches.

LEDs also have the advantage of being able to run on batteries, ushering in a wave of portable devices.

The Blue LED

In the early 1990s, the blue light LED was finally able to be produced with the consistent quality and quantities needed to ensure commercial success. Blue LEDs require a little more power to operate than their red and green cousins, but not so much they can't still run on batteries. In the late 1990s and early 2000s, blue LEDs began appearing in products everywhere.

With the blue LED also came the ability to create white-light LEDs. (White light is the combination of red, green, and blue light.) This paved the way for LED home lighting.

RGB LEDs

Soon after the blue LED came to market, RGB (red, green, blue) LEDs appeared. Instead of two leads, as all bulbs and LEDs had up to now, these LEDs had four leads, one for red, green, blue, and a common lead for a return path. Lighting them up in different combinations gave a wide range of colors, including white light.

Programmable RGB LEDs

As LEDs and digital electronics miniaturized, programmable LEDs began appearing in the early 2000s. Now, instead of variances in analog power to create different colors, digital signals could be used for precise control.

One popular programmable LED is the WS2812b, also known as a NeoPixel. This device, when paired with a small CPU/RAM/Flash memory combination called a microcontroller, can create different colors and intensities under computer control.

Addressable RGB LED Strips

If humans can control one RGB LED with a computer, it's only a matter of time until someone asks, "Hey, why not do a whole string of them?"

This is exactly what addressable RGB LED strips are. The same microcontroller that can control one can now control two or more, chained together. These LED strips have become popular in store displays and home holiday decorations. Some are even animated to music.

Organic LED Displays

Organic LED technology was also created in the 1990s and because of its efficiency, it allows dense packing of LEDs into things like computer displays. However, organic LEDs tend to wear out over time and suffer from an effect called "burn-in" where a ghost image can be seen in areas of dimmed output.

These OLED displays are often used in high-end computer and mobile phone displays, along with televisions. Their advantage of being brighter than traditional Liquid Crystal Displays (LCDs) makes them popular despite the drawback of burn-in.

Smaller, single color variants can be found in home appliances, like microwaves, electric ranges, washers, and dryers. These displays have the advantage of being able to show a wide variety of letters, numbers, and symbols, using simple programming.

Experiments

Below are some hands-on activities that tie in with each of the historical periods outlined above.

Tea Candle Lantern

Demonstration of fire as a light source. Use with caution and with a fire extinguisher nearby! Turn off the lights to see how well a single candle illuminates the room.

Battery and Flashlight Bulbs

Demonstration of old technology before white-light LEDs were popular. (This is what the kids of *Stranger Things* used to hunt supernatural monsters.)

For deeper exploration, use a multimeter set for resistance measurements to find about how many Ohms of resistance there are across one of the unused flashlight bulbs. Then measure the voltage across the battery in the circuit. Use this to calculate the current and power required for the flashlight bulb.

Red and Green LEDs

Create circuits on breadboards with a red and a green LED. Be sure to use 220 Ohm resistors to limit the current and not burn out the LEDs.

Use the multimeter's diode check feature to measure the voltage drop across an unused red and green LED. Compare the two numbers. Write down the order of the LEDs in terms of voltage drop.

For more advanced science, compare the LED order to a spectrum of light. Which color of light on the spectrum has the highest energy, red or green. How does this compare to the voltages measured?

Blue LEDs

Repeat the measurements used on the red and green LEDs with an unused blue LED. Put the blue LED in order with the red and green. Which had the highest voltage drop? How does this compare to the spectrum of light?

RGB LEDs

Use the multimeter's diode check to illuminate the various colors in the RGB LED. Place the RGB LED on a breadboard in such a way each color can be controlled individually. (Or use a microcontroller like Arduino to do this automatically.)

Point the LED toward a sheet of paper or other suitable "screen." Watch the circular pattern that is created for each color. Do they align? If not, why might that be?

Spoiler: there are actually three small individual LEDs inside. The fact they are competing for space means each one is separated from the other by a small bit. This difference is magnified when the LED is projecting onto a screen.

Programmable RGB LEDs

Create a MicroPython program with an ESP32-S3 SuperMini microcontroller to display some colors and combinations on the built-in WS2812b NeoPixel. Use individual color values ranging from 0 to 255.

For advanced math, calculate how many color combinations are theoretically possible with 256 values for each of the three colors.

Addressable LED Strips

Use an LED strip light combined with a microcontroller and WLED software to create a wirelessly controlled, animated light show on a strip of LEDs.

Compare this to the MicroPython code for the previous example.

Organic LED Displays

Wire an SSD1306-based OLED display to an ESP32-S3 SuperMini. Create a MicroPython program to display text messages on the display. For example, a program that prints the CPU and memory specifications of the microcontroller.

For an advanced activity, modify the MicroPython program, using the ESP32-S3's real-time clock to display time and date. (This will require network connectivity to accurately set the time.) If time synchronization is not available, simply have the display count upward in units of seconds since the device was powered on.